

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026

End-Semester Test
(EC-3 Regular)

Course No. : **SE ZG568 / SS ZG568**
Course Title : **Applied Machine Learning**
Nature of Exam : **Open Book**
Weightage : **40% (As per Course Handout)**
Duration : **2.5 Hours**
Date of Exam : **29/11/2025 (AN)**

No. of Pages	= 2
No. of Questions	= 5

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1

Marks 8

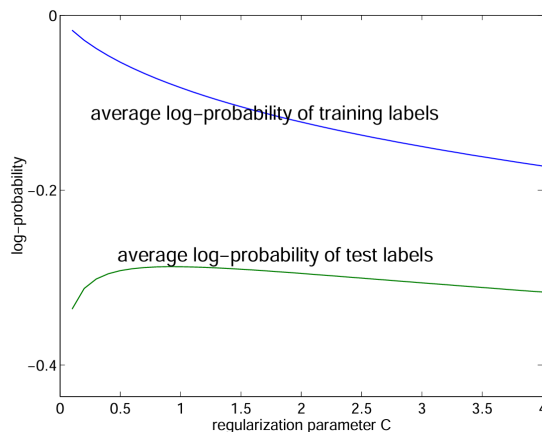
A	B	C	Y
F	F	F	F
T	F	T	T
T	T	F	T
T	T	T	F

Using the dataset above, we want to build a decision tree which classifies Y as T/F given the binary variables A, B and C. Draw the tree that would be learned by the greedy algorithm with zero training error.

Is this tree optimal (i.e. does it get zero training error with minimal depth)? Explain in detail. If it is not optimal, draw the optimal tree as well.

Q.2

Marks 10



Here, we use a logistic regression model to solve a classification problem. In the figure, we have plotted the mean log-probability of labels in the training and test sets after having trained the classifier with quadratic regularisation penalty and different values of the regularisation parameter C. In the context of logistic regression with regularisation, answer the following questions with an explanation:

- a. Can a logistic regression model maximising the likelihood of the labels given the inputs have multiple locally optimal solutions?
- b. Can the stochastic gradient algorithm for training logistic regression models with a fixed learning rate find the optimal setting of the weights exactly?

- Can the average log-probability of training labels presented in the figure increase as we increase the parameter C ?
- Explain why, in the figure, the test log-probability of labels decreases for large values of C .
- Will the log-probability of labels in the test set decrease for large values of C , even if we had a large number of training examples?

Q.3

Marks 10

Consider the following set of 3-dimensional points, sampled from two classes

	x_1	x_2	x_3		x_1	x_2	x_3
labeled '1':	1,	1,	-1	labeled '0':	1,	1,	2
	0,	2,	-2		0,	2,	1
	0,	-1,	1		1,	-1,	-1
	0,	-2,	2		1,	-2,	-2

Draw all possible 2-dimensional plots of pairs of features.

Assume that the learning is done with quadratic logistic regression, where:

$$P(y = 1|\mathbf{x}, \mathbf{w}) = g(w_0 + w_1x_i + w_2x_j + w_3x_ix_j + w_4x_i^2 + w_5x_j^2) \text{ for some pair of features } (x_i, x_j).$$

Based on the training set given above, which pair of features would result in the lowest training error for the logistic regression model? Explain your answer.

Q.4

Marks 6

Consider the following dataset, where '+' and 'O' represent the two classes.

O	+
+	O

- Which among the following models can successfully classify the dataset and why? (a) Logistic regression (b) linear regression (c) Depth-2 ID3 decision trees (d) Multilayer perceptron (e) 3 NN classifier
- If AdaBoost is used to classify the dataset, how will it handle the misclassified points?
- How will an SVM successfully classify the dataset?

Q.5

Marks 6

For the given data, compute two clusters using the K-means algorithm for clustering, where initial cluster centres are (1.0, 1.0) and (5.0, 7.0). Execute for one iteration and estimate the new centroids.

Record Number	A	B
R1	1.0	1.0
R2	1.5	2.0
R3	3.0	4.0
R4	5.0	7.0
R5	3.5	5.0
R6	4.5	5.0
R7	3.5	4.5